

Yousef A. Hassan

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OBJECTIVE

Computer Science graduate with practical software, data, and systems project experience using Python, C/C++, SQL, Linux/Unix, REST APIs, and clear technical documentation.

EDUCATION

Worcester State University

B.S. Computer Science; Minor in Physics; GPA: 3.06

Worcester, MA

May 2026

CLASS PROJECT EXPERIENCE

REST API Testing Suite [GitHub](#)

2025

HTTP / API Design

- Designed manual test cases for POST, PUT, and DELETE guest workflows, covering success paths and 400, 404, and 409 responses.
- Built request documentation for validation, duplicate-record checks, endpoint behavior, and automated API test planning.
- Organized expected inputs, request payloads, edge cases, and response outcomes into repeatable QA notes.
- Documented sample request bodies, expected status codes, and tester notes so API behavior was reviewed consistently.

Data Analytics Dashboard [GitHub](#)

2026

Python / SQL / Data Visualization

- Built a Python and SQL dashboard for cleaning a dataset, writing queries, generating summary metrics, and presenting trends in a simple web interface.
- Applied data mining, probability/statistics, databases, and capstone coursework to a documented analytics workflow.
- Prepared charts, calculated key metrics, and structured findings so non-technical users could scan results quickly.
- Compared cleaned outputs against summary calculations to catch inconsistent values before final visualization.

Linux Systems Toolkit [GitHub](#)

2026

Bash / UNIX Systems Programming

- Created Bash utilities for log parsing, file checks, permissions review, process monitoring, and repeatable command-line workflows.
- Documented each script with command examples to show Linux fundamentals, troubleshooting steps, and clean technical communication.
- Used UNIX systems programming concepts to make routine terminal tasks easier to test, explain, and repeat.
- Combined command-line flags, file-system checks, and script comments to support maintainable troubleshooting.

Database Design & Reporting Lab [GitHub](#)

2026

SQL / Schema Design

- Modeled a small business dataset with normalized tables, ERD notes, seed data, reporting queries, and clear tradeoff documentation.
- Built SQL queries, data quality checks, and practical reporting outputs that show database design beyond classroom exercises.
- Connected schema choices to reporting needs, validation rules, and maintainable data organization.
- Wrote reporting queries for joins, counts, filtered results, and grouped summaries tied to practical business questions.

Cloud Job Queue Demo [GitHub](#)

2026

Cloud / Distributed Systems

- Built a job-submission and worker-processing demo with status tracking, queue flow notes, and deployment documentation.
- Mapped cloud and distributed-computing concepts into task state, failure cases, and system visibility notes.
- Documented queue states such as submitted, processing, completed, and failed to make the workflow easier to inspect.

FarmData2 Repository Workflow [GitHub](#)

2025

JavaScript / Docker / Git

- Worked in a larger JavaScript and Docker codebase using Git branches, issue tracking, repository navigation, and pull-request workflow.
- Documented setup notes and codebase observations to show comfort contributing inside an existing project structure.
- Practiced reading unfamiliar modules, tracing setup requirements, and keeping changes organized for review.

Relevant Technical Coursework

2022-2026

Computer Science Coursework

- Applied coursework across Data Structures, Algorithm Analysis, Database Design & Applications, UNIX Systems Programming, Operating Systems, Computer Networking & Security, Data Mining, Big Data Analytics Capstone, Cloud/Parallel/Distributed Computing, Software Construction/Design/Architecture, Computer Organization & Architecture, Probability & Statistics, Technical Writing.
- Connected class projects to practical software, data, systems, documentation, and analysis workflows.
- Used Software Construction and Technical Writing coursework to organize project notes, testing steps, and implementation summaries.

SKILLS

Programming Languages: Python, C/C++, SQL, JavaScript, Bash/Shell

Systems & Tools: Linux/Unix, Git/GitHub, REST APIs, databases, cloud/distributed computing concepts, Netlify, Docker basics

Data & Documentation: Data mining, dashboard metrics, query design, technical writing, test documentation, workflow notes

AWARDS

Dean's List

Fall 2022, Spring 2026